

RAMITH K S

Senior Backend Engineer

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in ramith-k-s | ramithks

Bengaluru, India



PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years of experience building scalable Python/FastAPI services, distributed systems and cloud-native applications. Experienced in developing AI-enabled backend features using LangChain, LLM APIs, RAG pipelines and lightweight machine-learning models.

PROFESSIONAL EXPERIENCE

Thermo Fisher Scientific

Senior Backend Engineer

Feb 2026 – Present | Bengaluru, Karnataka, India (Hybrid)

E-Commerce checkout backend engineering via Kanmalai Technologies

- Build Python/FastAPI services for high-reliability e-commerce checkout and order-processing workflows on Kubernetes.
- Implemented Kafka-based acknowledgement processing, validation and status updates across asynchronous order workflows.
- Improved reliability through structured error handling, retries, observability and integration testing.

Zolnoi Innovations

Backend Engineer

Oct 2024 – Present | Bengaluru

Industrial IoT analytics company serving 50+ factories

- Built industrial IoT analytics APIs using FastAPI, PostgreSQL, Redis and MongoDB, processing 10,000+ daily requests.
- Integrated rule-based and ML-generated anomaly insights into production, energy and cycle-time analytics workflows.
- Designed a multi-channel alerting system with retries and failure handling, processing 5,000+ notifications daily.

JustPoll (Streetlab Internet Pvt Ltd)

Founder & CTO

Jul 2021 – Aug 2024 | Bengaluru

Social polling platform enabling opinion-based community engagement

- Founded and scaled a social platform to 1,500+ Android and 600+ iOS users; architected TypeScript backend and Flutter app.
- Boosted query performance by 40% via schema optimization; cut deployment time by 60% with ECS autoscaling and CI/CD pipelines.
- Led 5-member engineering team through 20+ production releases; improved user retention by 65% through mentoring and architecture reviews.

TECHNICAL SKILLS

Backend & Languages

Python, FastAPI, AsyncIO, REST APIs, Node.js, TypeScript

Generative AI & Machine Learning

LangChain, LangGraph, LLM API Integration, RAG, Prompt Engineering, Structured Outputs, Embeddings, scikit-learn, Model Serving

AI Data & Storage

pgvector, Vector Search, PostgreSQL, Redis, MongoDB, Pandas, NumPy

Systems & Cloud

Microservices, Kafka, Event-Driven Architecture, Kubernetes, Docker, AWS, GitHub Actions, CI/CD, Observability

Testing & Reliability

Pytest, Integration Testing, Retry and Fallback Workflows, Root Cause Analysis

Additional

React, Flutter

EDUCATION

Bachelor of Computer Applications

Christ University, Bengaluru

2018 – 2021 | Bengaluru

CGPA: 3.3 / 4.0

RECOGNITION

- ★ HackBangalore Winner 2024
- ★ AWS Cloud Foundations Certified
- ★ Open Source Contributor

PROJECTS

● AlgoHabit — AI-Powered Learning Platform

🌐 algohabit.com

- Built and launched AlgoHabit, an AI-powered DSA learning platform designed around habit formation, streaks, and gamified progression with 50+ active users.
- Engineered React PWA achieving 95+ Lighthouse performance score with offline-first architecture using service workers for 100% offline functionality.
- Built a LangChain-based LLM pipeline using OpenRouter, prompt templates, structured output parsing, retries and model fallback to generate personalised learning plans.
- Implemented a RAG pipeline using document chunking, embeddings and pgvector-based semantic search to generate syllabus-grounded learning recommendations.
- Deployed Supabase backend with Row-Level Security (RLS) securing 400+ user records; enabled zero-downtime Razorpay payments via serverless edge functions.
- Implemented gamification engine tracking 10,000+ XP transactions, improving 30-day retention by 60%.

● Ledgerly — Smart Finance Tracker SaaS

Personal Project

- Built scikit-learn pipelines using TF-IDF and Logistic Regression for transaction classification, statistical forecasting for spending trends and MAD-based anomaly detection.
- Served ML predictions through FastAPI endpoints with Pydantic validation, model loading, logging and health checks.
- Used MLflow for basic experiment tracking, model metrics and version comparison.

● HelpMyBuddy — HackBangalore Winner 2024

🐙 github.com/SiliconNinjas

- Built real-time geo-based emergency help marketplace connecting users to nearby helpers for breakdowns and medical emergencies.
- Designed a low-latency geospatial matching engine using MongoDB spatial queries, achieving <500ms matching within a 5 km radius.
- Developed React Native (Expo) mobile app enabling live location sharing, helper discovery, and emergency request workflows.
- Built TypeScript-based Node.js + Express backend to handle authentication, matching, and request orchestration with FastAPI-powered ML services.

● AG-Wallet — Agricultural Supply Chain Platform

2020 – 2021

- Built Android app connecting farmers to distributors with real-time price tracking and downstream visibility.
- Integrated Google Maps API for location-based discovery and rating system for distributor evaluation.
- Enabled transparent agricultural supply chain with location-based discovery and quality ratings.